UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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QUESTION 1

a) Write the solution of 5x - 7 > 1 in interval notation and on a number line.

(3 marks)



Figure: solution set.

b) Solve the inequality |2x + 5| >= 6. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Sketch y = 2|x + 1| + 2. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

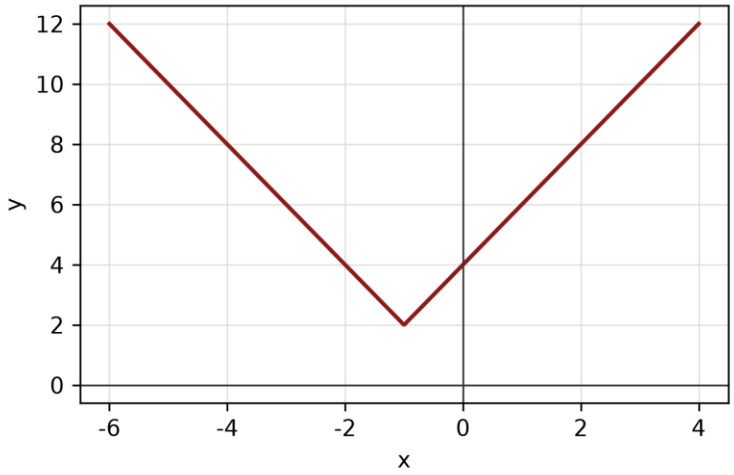


Figure: absolute-value graph.

QUESTION 2

a) Apply graph-transformation techniques to sketch $y = -3(x - 2)^2 - 3$. State the vertex and the y-intercept.

(5 marks)

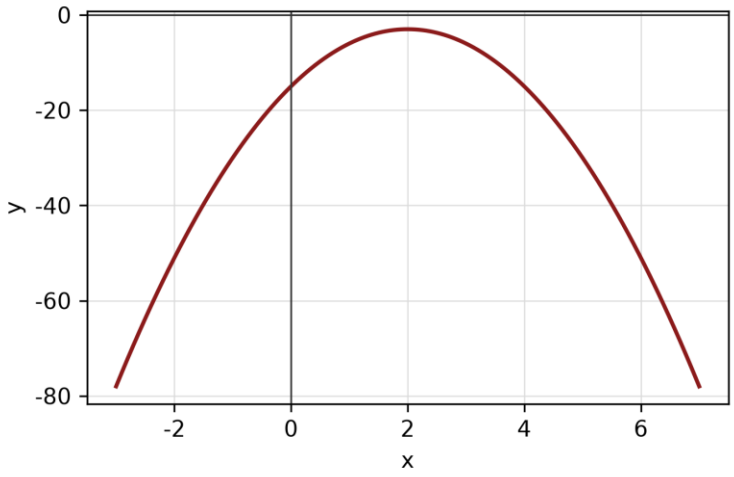


Figure: transformed graph.

b) Sketch $y = \sqrt{(x + 2)}$ for $-2 \leq x \leq 7$. State the domain and range.

(4 marks)

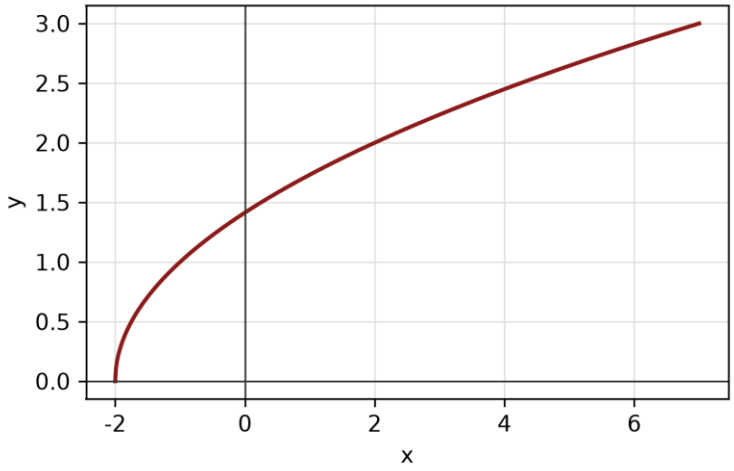


Figure: function graph.

c) Find the equation of the line passing through (2, 3) and perpendicular to the line -5x - y - 6 = 0.

(5 marks)

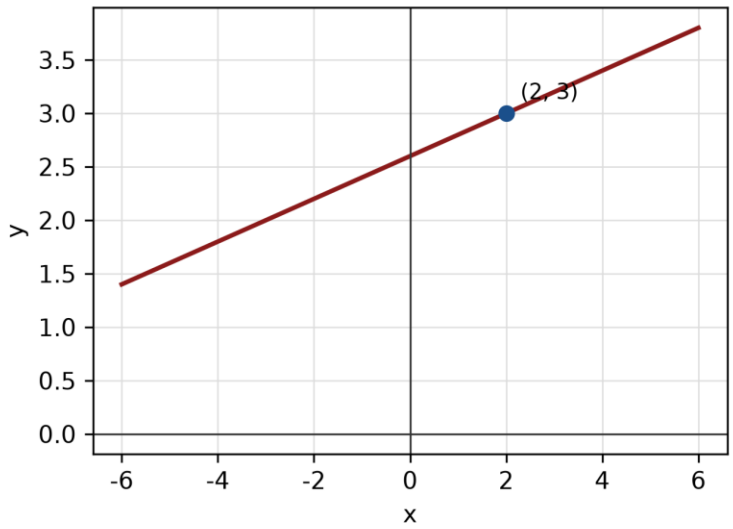


Figure: required line.

QUESTION 3

a) Given $z_1 = 4 - 3i$ and $z_2 = -2 + 2i$, express z_1/z_2 in the form a + bi.

(5 marks)

b) Find the modulus and argument of z = -5 + 12i. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Show that x + 3 is a factor of f(x) = 3x^3 + 6x² - 13x - 12 by using the factor theorem. Hence factorise f(x) completely.

(5 marks)

QUESTION 4

a) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 1x + 2y \leq 6.$

(5 marks)

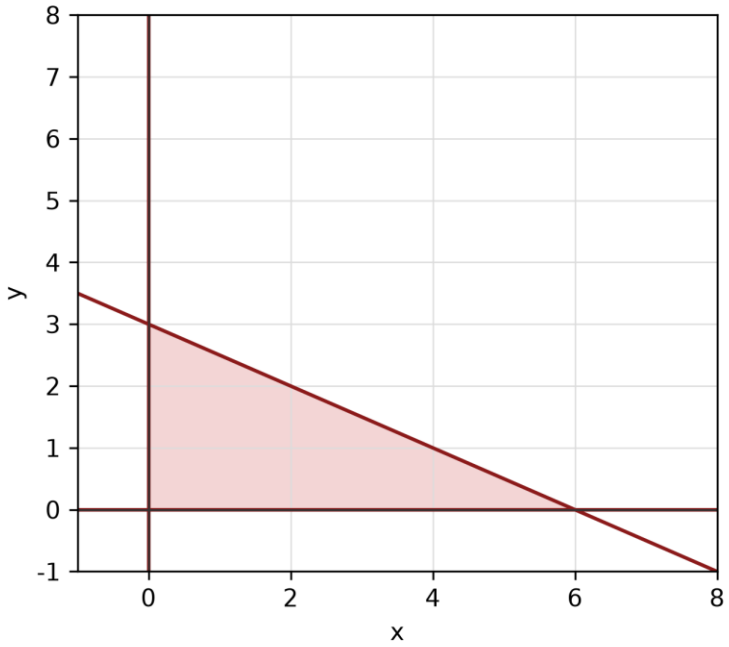


Figure: feasible region.

b) Solve the simultaneous equations
$y = x^2 - 2$
$y = 7x - 12.$

(5 marks)

c) Write $y = x^2/8 + 3x/4 + 49/8$ in standard form. Hence state the vertex, focus and directrix, and sketch the parabola.

(5 marks)

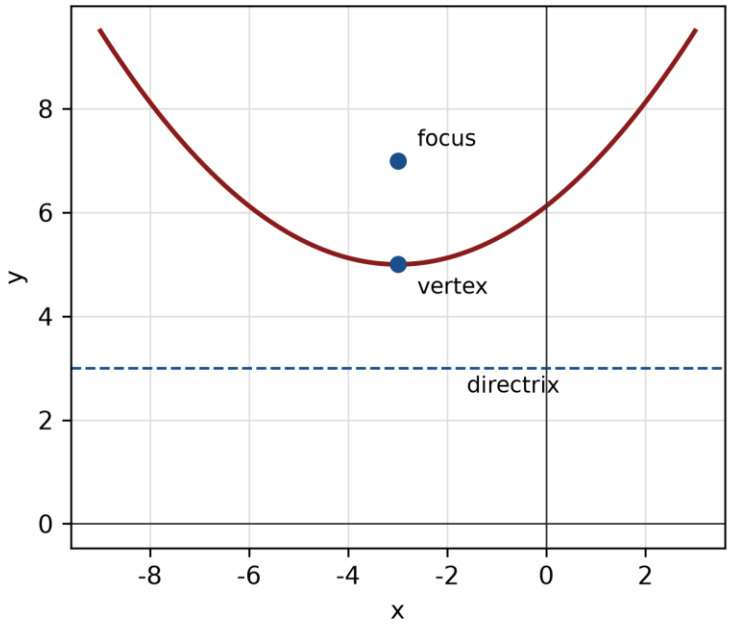


Figure: parabola.

QUESTION 5

a) Given sin A = 8/17 where A is in quadrant II and cos B = 15/17 where B is in quadrant IV, find sin 2A and cos(A + B).

(5 marks)

b) Find the perimeter of the sector with radius 15 cm and central angle 90°.

(4 marks)

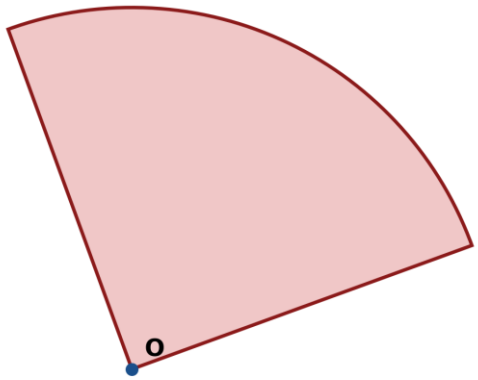


Figure: sector.

c) In triangle ABC, b = 10 cm, c = 11 cm and A = 120°. Use the cosine rule to find a.

(4 marks)

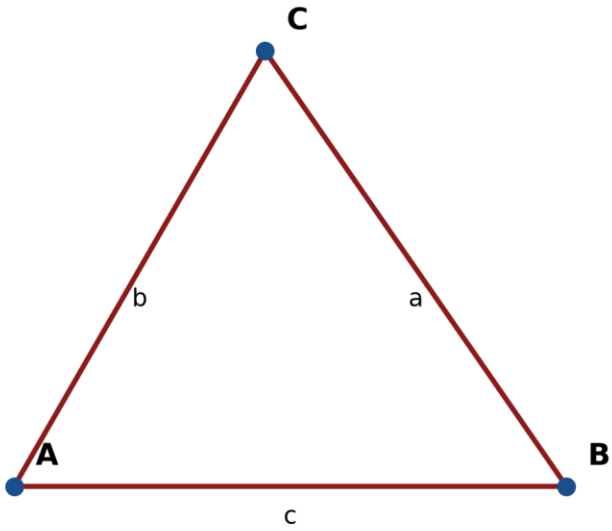


Figure: triangle ABC.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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